

The Megarians and the Stoics

From Dialectical Paradox to Propositional Logic

History of Logic

Primary source: William and Martha Kneale, *The Development of Logic*, Chapter III, “The Megarians and the Stoics.”

1 Two Great Traditions of Ancient Logic

The preceding chapter concentrated upon Aristotle.

Aristotle’s mature formal theory grew chiefly out of reflection upon reasoning involving general terms. Geometry and the ideal of demonstration provided an especially important model.

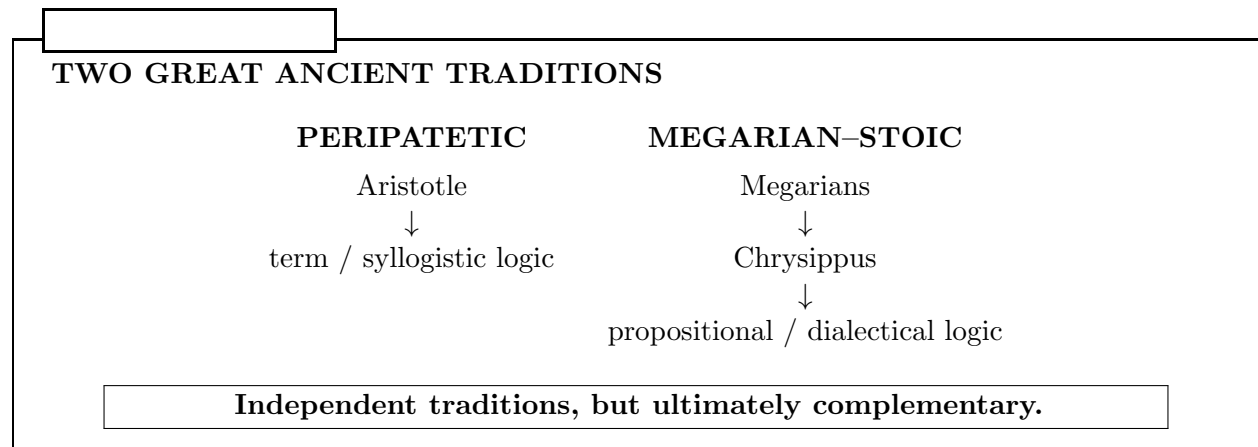
But Aristotle’s was not the only great logical tradition of the ancient world.

Later antiquity distinguished two major schools:

1. the **Peripatetic** tradition deriving from Aristotle;
2. the **Stoic** tradition, developed especially by Chrysippus from Megarian logical work.

The difference between the two traditions is not simply a difference of doctrine.

They begin from different kinds of argumentative practice and therefore isolate different kinds of logical structure.



Unfortunately, the writings of the great Stoic logicians have not survived intact.

Our knowledge comes primarily from later reports, often written by people who belonged to rival philosophical schools.

We can reconstruct the major features of Stoic logic, but the evidence is sometimes incomplete or uncertain.

2 Geometry and Dialectic: Two Sources of Logic

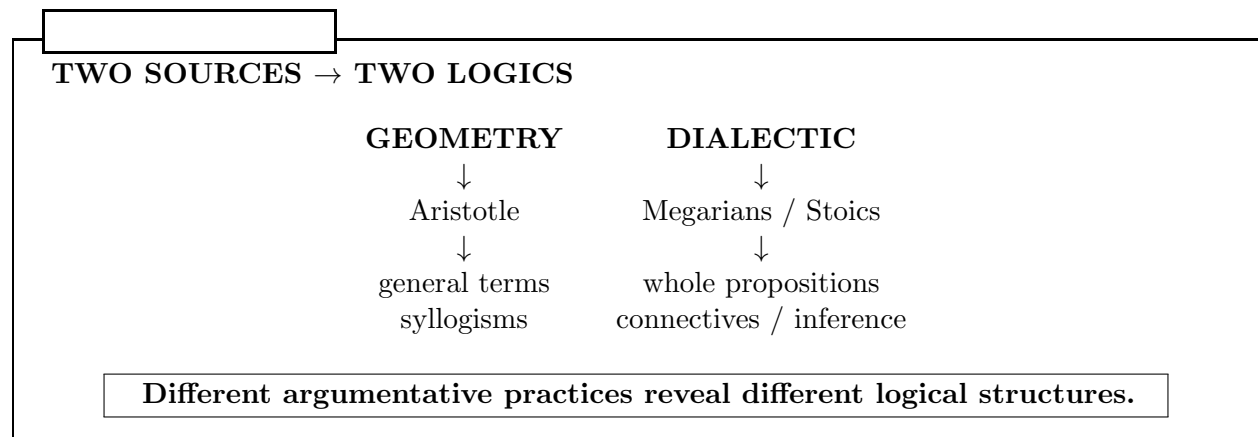
The contrast between Aristotle and the Megarians can be traced back to the kinds of reasoning upon which they concentrated.

Aristotle appears to have been strongly influenced by demonstrative reasoning of the sort found in mathematics.

The Megarians, by contrast, inherited traditions associated with:

- Eleatic philosophy;
- Zeno's dialectical arguments;
- reductio;
- eristic disputation;
- paradox;
- conditional argument.

These different origins help explain why Aristotle and the Stoics developed such different formal systems.



This contrast will organize the entire lecture.

I. The Megarian Tradition and Its Paradoxes

3 From Euclides to Chrysippus

Euclides of Megara, an older contemporary of Plato, is traditionally regarded as the founder of the Megarian school.

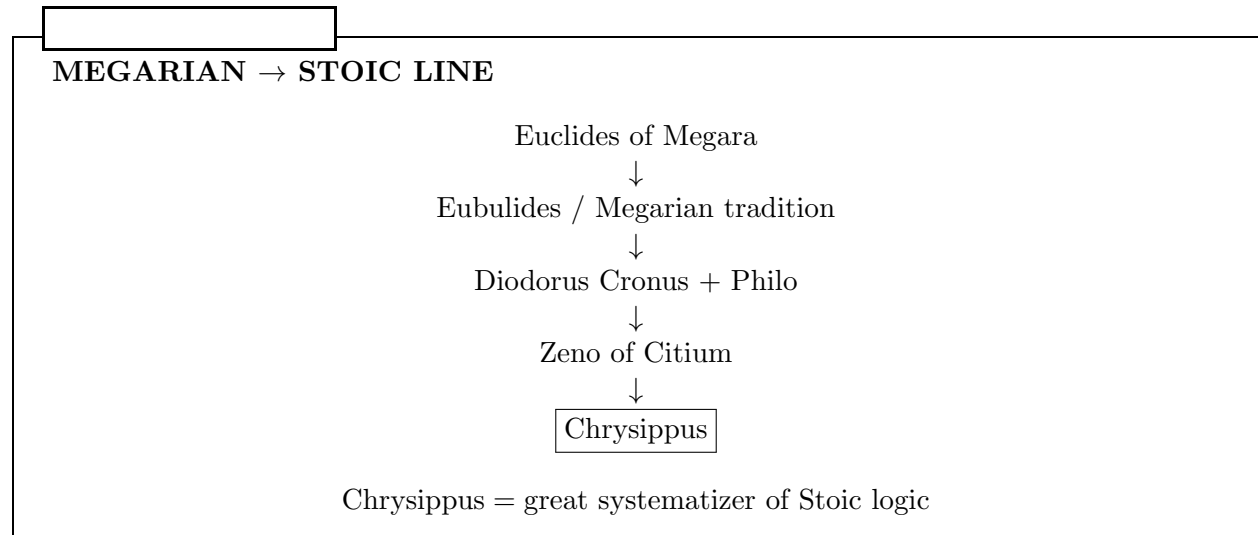
The line of influence is not perfectly recoverable, but several figures are especially important.

Eubulides became famous for paradoxes.

Diodorus Cronus and Philo developed theories of modality and conditionals.

Stilpo belonged to the Megarian tradition and taught Zeno of Citium, the founder of Stoicism.

Chrysippus, the third major head of the Stoic school, eventually gave Stoic dialectic its most systematic form.



Ancient admiration for Chrysippus as a dialectician was extraordinary.

His logical work was regarded as a rival to Aristotle's system, not merely as a minor supplement to it.

4 Eubulides and the Logical Use of Paradox

Eubulides was traditionally credited with several paradoxes.

The Kneales reduce the traditional list to four principal types:

1. the Liar;
2. the Hooded Man, also called the Unnoticed Man or Electra;
3. the Bald Man or Heap;
4. the Horned Man.

These puzzles are important because each exposes a genuine logical or philosophical problem.

They should not be dismissed as mere verbal games.

EUBULIDES: FOUR PARADOXES

1. Liar
2. Hooded Man / Electra
3. Bald Man / Heap
4. Horned Man

Paradox becomes a tool for discovering logical problems.

5 The Liar

The Liar concerns a statement that turns upon its own truth or falsity.

A modernized compact form is:

“This statement is false.”

Suppose the statement is true.

Then what it says is the case, and therefore it is false.

But suppose it is false.

Then what it says is not the case, which seems to make it true.

THE LIAR

“This statement is false.”

If TRUE:

what it says $\Rightarrow$ FALSE

If FALSE:

what it says appears $\Rightarrow$ TRUE

Problem: SELF-REFERENCE + TRUTH

The purpose of this lecture is not to solve the Liar.

Its historical importance lies in the fact that Megarian dialecticians had isolated a profound difficulty involving self-reference and the predicates “true” and “false.”

6 The Hooded Man

The Hooded Man raises a different problem.

Suppose you know your brother.

A hooded person enters.

Unknown to you, the hooded person is your brother.

You do not recognize the hooded person.

Must we therefore say both:

you know your brother

and:

you do not know your brother?

The puzzle concerns the interaction of identity with contexts involving knowledge.

HOODED MAN

Suppose:

$$X = Y$$

Question:

Can everything said truly of X be substituted for Y ?

Identity alone may not preserve truth in knowledge-contexts.

The Kneales also note that the puzzle forces us to distinguish different uses of the word “know.”

We need not yet develop a full theory of such contexts.

The important point is that simple substitution of identical objects can become problematic when the proposition also describes someone’s knowledge.

7 The Heap and the Bald Man

The Heap and the Bald Man are versions of what later came to be called the sorites paradox.

Suppose a man with only one hair is bald.

Would two hairs make him non-bald?

Three?

Four?

At what precise point does the change occur?

The heap version asks the corresponding question about grains of material.

THE HEAP / BALD MAN

1 hair $\rightarrow$ bald?

2 hairs $\rightarrow$ bald?

3 hairs $\rightarrow$ bald?

 $\vdots$

WHERE IS THE BOUNDARY?

Problem: VAGUE PREDICATES

The paradox reveals that many ordinary predicates do not possess sharp boundaries of application.

8 The Horned Man

The Horned Man takes the form:

What you have not lost, you still have.

You have not lost horns.

Therefore:

You still have horns.

The argument is fallacious, but the interesting question is why.

The Kneales interpret the difficulty in terms of **presupposition**.

The statement:

“You have lost your horns”

normally presupposes that you once had horns.

But a denial can operate in more than one way.

It may reject only the explicit claim that you lost them while leaving the presupposition intact, or it may reject the entire complex claim.

HORNED MAN

What you have not lost, you have.

You have not lost horns.

∴ You have horns.

Hidden issue:

“You lost horns”

presupposes:

“You once had horns.”

Negation may reject more than the explicit predicate.

9 What the Four Paradoxes Reveal

The puzzles attributed to Eubulides anticipate four enormous areas of later logical inquiry.

PARADOX → LOGICAL PROBLEM

Liar → truth / self-reference

Hooded Man → identity / knowledge

Heap → vagueness

Horned Man → presupposition / negation

Whatever role entertainment and controversy played in Megarian practice, the intellectual content of these puzzles is serious.

10 A Historical Mistake: Rivalry Instead of Combination

The Megarian and Stoic schools became involved in a long rivalry with the Peripatetic tradition.

The Kneales regard this as unfortunate.

Aristotelian and Stoic logic are not naturally alternatives.

They study different dimensions of logical structure.

ANCIENT POLEMIC

Aristotle **VS.** Stoics

Better picture:

Aristotle + Stoics

TERM LOGIC + PROPOSITIONAL LOGIC

The failure to integrate these two achievements will become one of the chapter's central historical conclusions.

II. Megarian and Stoic Theories of Modality

11 The Problem of Possibility

Aristotle reports that some Megarians denied a genuine distinction between actuality and potentiality.

Such a doctrine fits naturally with an Eleatic suspicion of motion and change.

Yet later Megarians devoted substantial attention to modal notions such as:

- possible;
- impossible;
- necessary;
- non-necessary.

The question becomes:

Can modality be explained without Aristotle's theory of real potentiality?

THE MODAL QUESTION

What do we mean by:

POSSIBLE
IMPOSSIBLE
NECESSARY
NON-NECESSARY

Diodorus $\neq$ Philo $\neq$ later Stoics

12 Diodorus Cronus: Modal Definitions

Diodorus offers the most striking of the Megarian modal theories.

He defines modality through the truth or falsity of a statement across time.

Using P as a neutral placeholder for the relevant statement:

DIODORUS

Possible $P = P$ is true or will be true

Impossible $P = P$ is false and will never be true

Necessary $P = P$ is true and will never be false

Non-necessary $P = P$ is false or will be false

The modal predicates are thus tied directly to patterns of truth through time.

13 Diodoran Modality Is Modality-at-a-Time

The Kneales emphasize that Diodorus is not defining necessity simply and timelessly.

He is effectively defining **necessity-at-a-time**.

Suppose a statement is now true and will never again become false.

It is now necessary according to Diodorus.

But before the relevant event occurred, the corresponding statement may merely have been possible.

Thus modality can change.

DIODORAN MODALITY

modality depends upon truth through TIME

Possible changes:

possible $\rightarrow$ necessary

possible $\rightarrow$ impossible

But once reached:

necessary $\rightarrow$ stays necessary

impossible $\rightarrow$ stays impossible

The theory is closely connected with Diodorus' famous **Master Argument**.

14 The Master Argument

The Master Argument is known chiefly through later reports.

It begins with three propositions which Diodorus claims are mutually incompatible.

1. Everything that is past and true is necessary.
2. The impossible does not follow from the possible.
3. Something which neither is nor will be true may nevertheless be possible.

Diodorus accepts the first two and rejects the third.

Hence he identifies the possible with what either is or will be true.

THE MASTER ARGUMENT

Three claims cannot all stand:

1. Past truth $\rightarrow$ necessary
2. Impossible cannot follow from possible
3. Something may be possible
although it never occurs

Diodorus:

accept 1 + 2

	reject 3	
Therefore:		
	Possible = what is or will be true	

This conclusion was naturally associated in antiquity with determinism or fatalism.

15 Why the Master Argument Seems Plausible

The first premise gains much of its force from a familiar thought:

What has happened cannot now be made not to have happened.

Even Aristotle remarks that deliberation concerns the future, not the already completed past.

But a distinction is required.

The past may now be **unalterable** without thereby having been **absolutely necessary** before it occurred.

	WHY PREMISE 1 TEMPTS US	
	What has happened	
	↓	
	cannot now be changed	
Temptation:		
	cannot now be changed $\Rightarrow$ was absolutely necessary	
But:		
	UNALTERABILITY $\neq$ ABSOLUTE NECESSITY	

This is closely related to the distinction between relative and absolute necessity encountered in Aristotle.

16 The Ambiguity in the Master Argument

The Kneales identify a crucial ambiguity.

We must distinguish:

1. a statement **about the past**;
2. a statement grammatically **in the past tense**.

These are not equivalent.

A sentence can be in the past tense while its interpretation depends upon the present context of utterance.

Likewise, a sentence with a different tense can make a claim about an event already past.

CRUCIAL DISTINCTION

statement ABOUT the past $\neq$ statement IN the past tense

Underlying problem:

sentence $\neq$ what the sentence expresses

The Kneales argue that the Master Argument trades upon this ambiguity.

The problem resembles one already encountered in Aristotle's discussion of future contingents.

Both theories suffer from insufficiently sharp separation of sentences from the propositions or contents expressed by them.

17 Philo's Alternative Theory of Possibility

Philo rejects Diodorus' restriction of possibility to what does or will actually occur.

For Philo, something can be possible in virtue of the intrinsic character of the subject even when external conditions prevent its realization.

The ancient example concerns wood at the bottom of the sea.

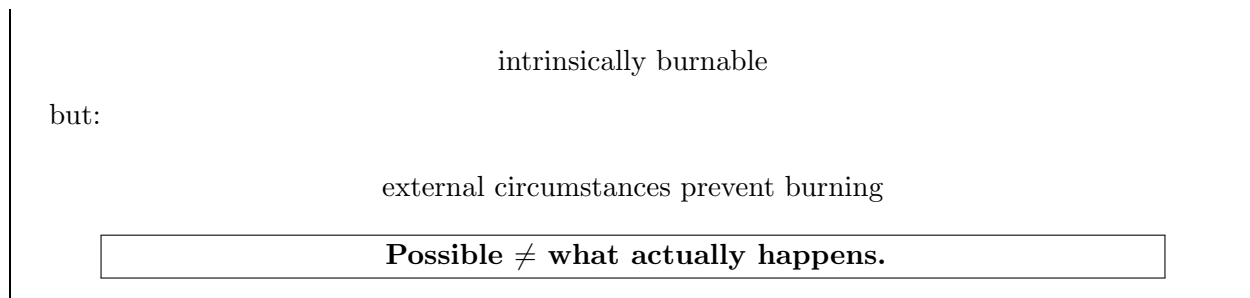
The wood remains the sort of thing that can burn even though its present surroundings prevent it from burning.

PHILO

Possible = admits of truth by its intrinsic nature

Example:

wood under the sea



The Kneales interpret Philo as making possibility the basic modal notion and associating it approximately with intrinsic consistency or admissibility.

18 The Later Stoic Variant

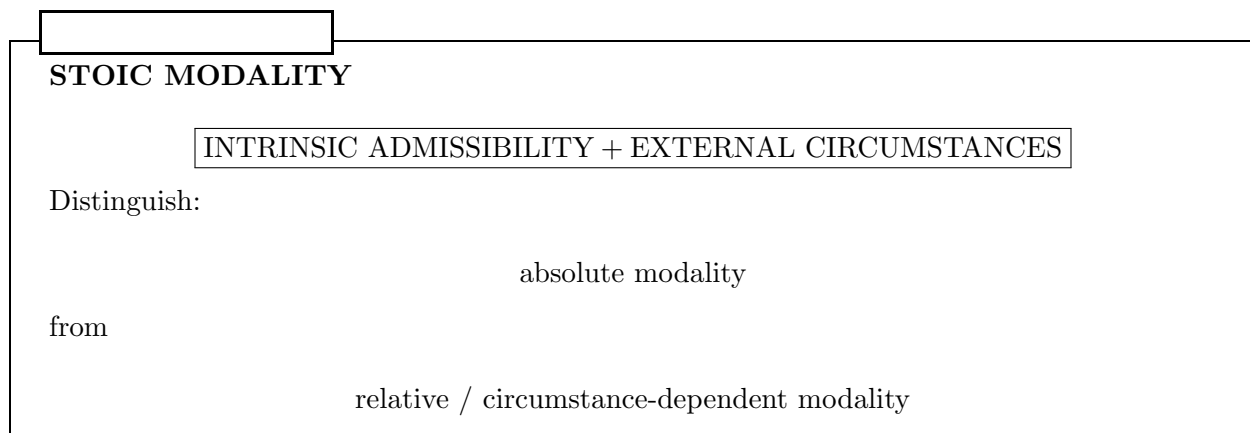
Later reports attribute to the Stoics a more complicated theory.

The evidence is compressed and does not permit complete certainty.

But the important distinction appears to be between:

1. what something intrinsically admits;
2. what external circumstances allow or prevent.

Thus modal status can involve both intrinsic and circumstantial features.



Because our ancient evidence is fragmentary, this reconstruction must remain cautious.

19 Three Ancient Strategies for Modality

The differences can now be summarized.

THREE APPROACHES

Diodorus → truth through time
Philo → intrinsic admissibility
Stoics → intrinsic admissibility
+ external circumstances

Same modal vocabulary; different underlying theories.

The modal debate prepares directly for another Megarian controversy: the nature of conditional statements.

III. The Debate on the Nature of Conditionals

20 Why Dialecticians Care About “If”

Conditionals were not central to Aristotle’s classification of propositions.

For a thinker working in the tradition of Zeno, however, conditional reasoning is unavoidable.

A typical refutation has the following shape:

$$P \rightarrow Q,$$

$$P \rightarrow \neg Q,$$

therefore:

$$\neg P.$$

If arguments of this sort are fundamental to dialectical practice, the question immediately arises:

What exactly does “if P , then Q ” mean?

ZENONIAN DIALECTIC

$$P \rightarrow Q$$

$$P \rightarrow \neg Q$$

$$\therefore \neg P$$

WHAT DOES $P \rightarrow Q$ MEAN?

The first extensive ancient debate over this problem appears among Diodorus, Philo, and later dialecticians.

21 Four Ancient Theories of the Conditional

Sextus Empiricus reports four views.

The authorship of all four is not certain.

Philo and Diodorus are securely identified with the first two.

The third may well be associated with Chrysippus.

FOUR THEORIES OF “IF P , THEN Q ”

1. **Philo:** not (P true and Q false)
2. **Diodorus:** never could / can be
 P true and Q false
3. **Connexion:** P incompatible with $\neg Q$
4. **Implication:** Q contained potentially in P

The disagreement concerns how strong a connexion must obtain between antecedent and consequent.

22 Philo’s Truth-Functional Conditional

Philo gives the weakest of these theories.

A conditional is false only when its antecedent is true and its consequent false.

In every other combination of truth-values, the conditional is true.

Using modern tabular notation:

PHILO’S CONDITIONAL

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Only false case: $P = T, Q = F$

The truth-table device itself is modern.

The important historical fact is that the idea of **truth-functional dependence** is already clearly present.

23 Why Philo’s Theory Makes Logical Sense

At first sight Philo’s theory gives strange results.

Why should:

$$P \rightarrow Q$$

be true merely because P is false?

The Kneales suggest that Philo may have begun from a fundamental property required of conditional reasoning.

Whatever else “if P , then Q ” means, it must license the following pattern:

MINIMUM REQUIREMENT FOR “IF”

$$\frac{P \rightarrow Q \quad P}{Q}$$

Philo gives the weakest truth-condition

that guarantees this inference.

This inference will later become the first of Chrysippus’ five basic inference schemata.

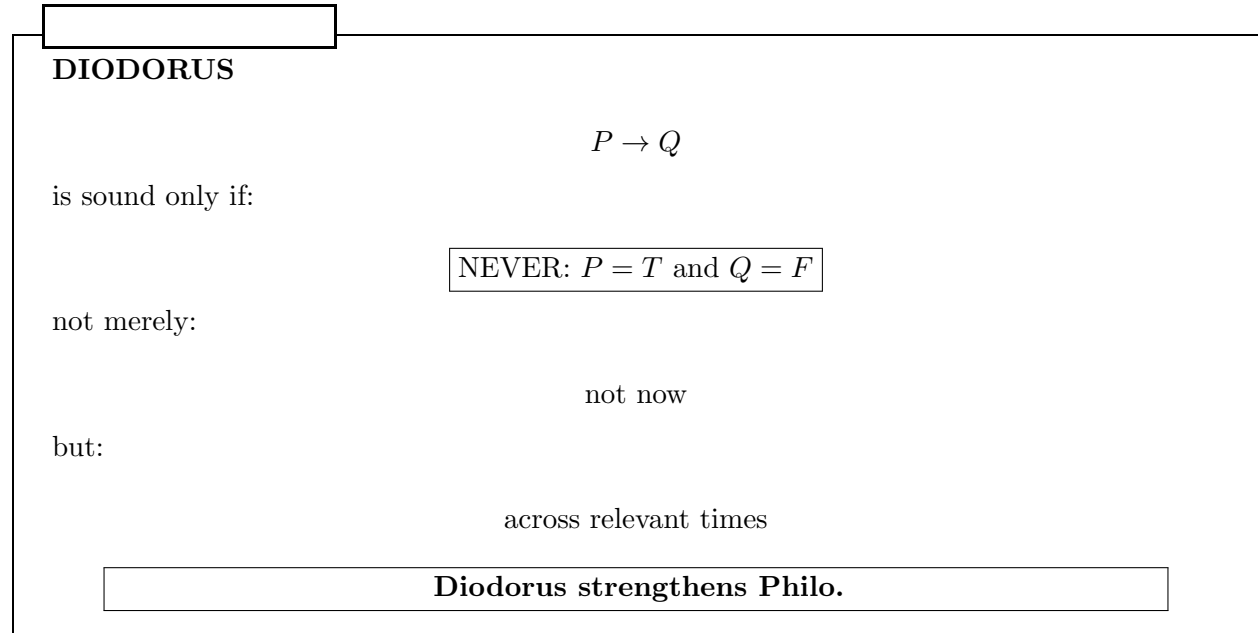
24 Diodorus’ Stronger Conditional

Diodorus rejects Philo’s account as too weak.

A conditional should not count as sound merely because its antecedent happens now to be false or its consequent happens now to be true.

Diodorus therefore adds a temporal requirement.

A sound conditional must never have been and never be capable of beginning with a truth and ending with a falsehood.



This theory is closely tied to Diodorus' peculiar treatment of modality and temporal truth.

It therefore inherits many of the difficulties of that theory.

25 The Theory of Connexion

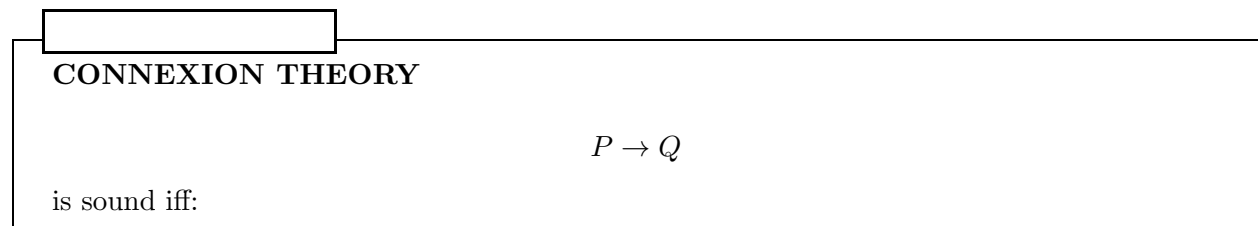
The third reported theory says that a sound conditional must contain a stronger connexion.

The negation of the consequent must be incompatible with the antecedent.

Thus:

$P \rightarrow Q$

is sound when P cannot hold together with $\neg Q$.



That is:	P is incompatible with $\neg Q$	
not possible: $P \wedge \neg Q$		

The surviving evidence suggests that this may have been Chrysippus’ characteristic position, but the attribution is not certain enough to state without qualification.

26 Conditionality and Entailment

The debate exposes an important distinction.

If P logically entails Q , then we may certainly form the true conditional:

$$P \rightarrow Q.$$

But the converse does not automatically hold.

Not every conditional accepted as true in ordinary discourse is therefore an expression of logical entailment.

CONDITIONALITY VS. ENTAILMENT	
	$P \models Q$
supports:	
	$P \rightarrow Q$
Therefore:	
	ENTAILMENT $\Rightarrow$ true conditional
But:	
	true conditional $\nRightarrow$ logical entailment
CONDITIONALITY $\neq$ ENTAILMENT	

Much later confusion in logic resulted from attempts to identify all uses of “if ... then ...” with necessary logical connexion.

27 The Kneales on Primitive Conditional Speech

At this point the Kneales move from historical reconstruction to their own analysis of ordinary conditional speech.

This should be distinguished clearly from the doctrines of Philo, Diodorus, or Chrysippus.

Consider a promise:

“I will do Q if P occurs.”

If P never occurs, we do not ordinarily say that the speaker has fulfilled the promise merely because the antecedent is false.

Similarly, a conditional order does not count as obeyed simply because the relevant condition never occurs.

The Kneales suggest that one primitive function of an “if”-clause is to select a possibility for attention.

What occurs in the consequent is then said with respect to that possibility.

KNEALES: PRIMITIVE “IF”

If P , then Q

Interpretation:

$P \rightarrow$ selects a POSSIBILITY

$Q \rightarrow$ said with respect to that possibility

If P never occurs:

the conditional may have NO APPLICATION

28 The Defective Truth-Function

On this analysis, a primitive conditional can be represented as a **defective truth-function**.

When the antecedent is fulfilled, the truth or falsity of the consequent determines whether the conditional prediction was borne out.

When the antecedent is false, the conditional may simply have no application.

DEFECTIVE TRUTH-FUNCTION

P	Q	If P then Q
T	T	T
T	F	F
F	T	—
F	F	—

— = no application

Not automatically bivalent.

Again, this is the Kneales' philosophical analysis, not an ancient Stoic truth-table.

29 Necessary-Connexion Conditionals

Other conditional statements do something stronger.

The speaker may mean:

P cannot occur without Q .

Such a claim is not established merely by discovering that P and Q happen both to be true.

It asserts some modal connexion between them.

TWO IMPORTANT USES OF “IF”

Primitive: If P , expect Q .

Modal: P cannot hold without Q .

truth-value dependence $\neq$ necessary connexion

No single simple rule captures every ordinary use of “if.”

The historical importance of the Megarian debate lies partly in the fact that ancient logicians already recognized the difficulty of explaining conditional connexion.

IV. Stoic Meaning, Truth, and the Lekton

30 Logic as a Part of Philosophy

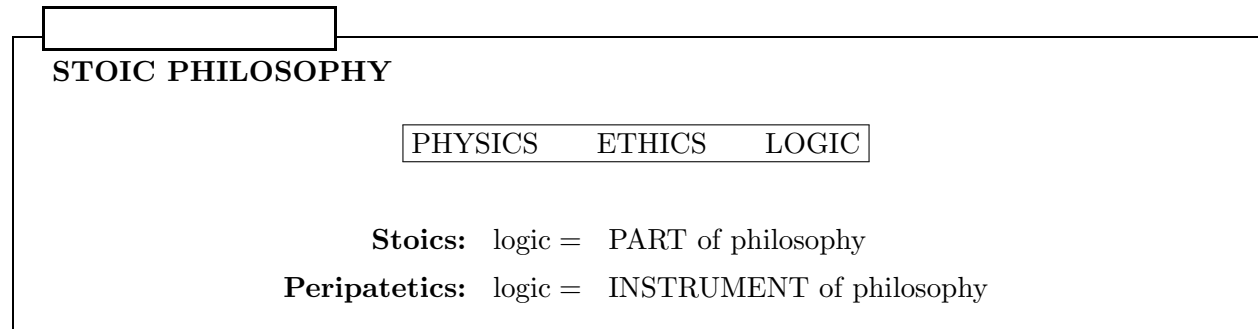
The Stoics gave logic a more explicit place in their philosophical system than any of their predecessors.

Their standard division of philosophy was:

1. physics;
2. ethics;
3. logic.

The Peripatetics were later represented as treating logic mainly as an **instrument** of philosophy.

The Stoics treated it as one of philosophy's constituent parts.



Within logic, the Stoic term **dialectic** corresponds most closely to what we now call logic, though Stoic dialectic also includes material we would place under grammar, semantics, and epistemology.

31 Things Signifying and Things Signified

The Stoics divided dialectic into investigation of:

- things that signify;
- things that are signified.

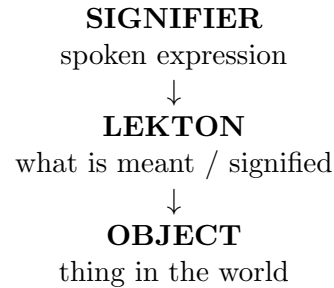
They developed a remarkably sophisticated theory of linguistic meaning.

The central distinction involves three items:

1. the signifying expression;
2. what is signified or meant;
3. the external object.

What is signified or meant is called a **lekton**.

STOIC SEMANTIC TRIAD



In Stoic metaphysics:

- speech is corporeal;
- the external object is corporeal;
- the lekton is incorporeal.

speech	=	corporeal
object	=	corporeal
lekton	=	INCORPOREAL

This is striking because the Stoics were otherwise strongly materialist philosophers.

32 Why the Lekton Is Logically Important

The theory responds to a problem that has appeared repeatedly throughout our course.

What exactly is true or false?

A physical sound cannot by itself be the primary bearer of truth.

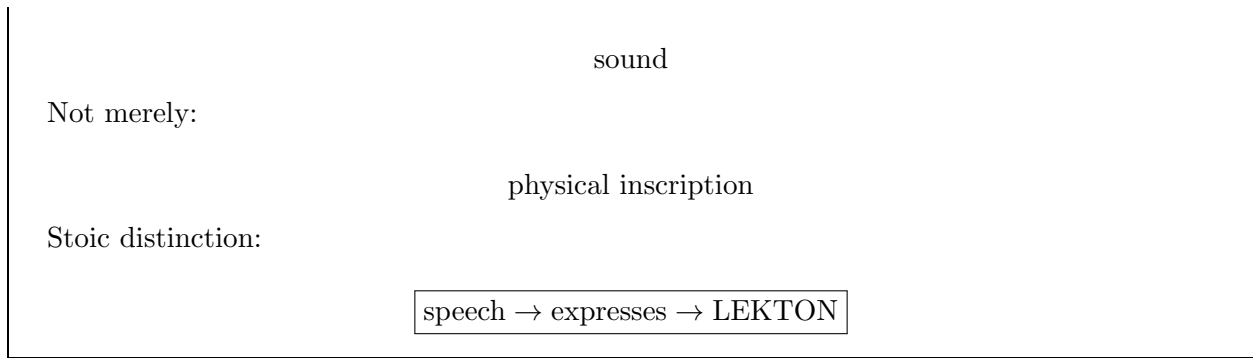
Someone who does not understand Greek can hear a Greek sentence without understanding what it signifies.

The Stoics therefore distinguish the expression from what the expression means.

OLD PROBLEM

What is TRUE or FALSE?

Not merely:



But we must not immediately identify a lekton with a modern proposition.

The Stoic theory is more complicated.

33 Complete and Defective Lekta

The Stoics divide lekta into complete and defective.

A defective lekton does not by itself constitute a finished content.

For example:

“writes”

leaves us asking:

Who writes?

By contrast:

“Socrates writes”

expresses a complete lekton.

LEKTA	
	DEFECTIVE: “... writes” $\Rightarrow$ Who?
	COMPLETE: “Socrates writes.”
	predicate alone $\rightarrow$ incomplete
	finished content $\rightarrow$ complete

Defective lekta include such things as predicates and subjects considered apart from complete assertion.

34 Axiomata

Among complete lekta, the most important logical class is the **axioma**.

The word must not be confused with the modern mathematical word “axiom.”

A Stoic axioma is roughly a complete declarative content capable of truth or falsity.

Other complete lekta include:

- questions;
- commands;
- prayers;
- oaths;
- suppositions;
- addresses.

These need not themselves be true or false.

COMPLETE LEKTA

AXIOMA → TRUE or FALSE

question → not true / false

command → not true / false

prayer → not true / false

AXIOMA = complete declaratory lekton capable of truth or falsity.

This is a significant advance in the analysis of meaning.

Not everything that has complete meaning is therefore a bearer of truth-value.

35 Where Truth and Falsity Belong

The Stoics explicitly distinguish several candidates for truth and falsity.

According to the doctrine reported by Sextus, the fundamental bearer of truth is neither:

- the spoken expression as such;
- nor merely a physical or psychological movement of thought.

Truth and falsity fundamentally concern what is signified.

For logical purposes, the axiom becomes the crucial bearer of truth-value.

WHAT IS TRUE OR FALSE?

Not fundamentally:

sound / sentence-token

Not fundamentally:

mental movement

Stoic answer:

AXIOM / SIGNIFIED CONTENT

truth is fundamentally attached to what is meant

The Stoics also used “true” and “false” in derivative ways for presentations and arguments.

A true argument, for example, was valid and had true premisses.

36 The True and the Truth

The Stoics even distinguish:

the true

from:

truth.

The true is an incorporeal axiom.

Truth, in the special Stoic sense, is a body of knowledge possessed by a knower.

This distinction belongs partly to Stoic metaphysics and epistemology rather than formal logic, so we need only record its structure.

STOIC DISTINCTION

the true = a true axiom
incorporeal

truth = a body of knowledge
corporeal in Stoic theory

Our logical concern remains chiefly with true and false axiomata.

37 Negation and the Discovery of Scope

The Stoics give careful attention to negation.

To form the contradictory of an axiomata, the negative must govern the **whole** axiomata.

Compare:

$$\neg(P \wedge Q)$$

with:

$$P \wedge \neg Q.$$

These are not the same proposition.

In the first, negation applies to the entire conjunction.

In the second, it applies only to Q .

SCOPE OF NEGATION

$$\neg(P \wedge Q)$$

means:

NOT: both P and Q

But:

$$P \wedge \neg Q$$

means:

P and not- Q

Therefore:

$$\neg(P \wedge Q) \neq P \wedge \neg Q$$

Negation of the whole $\neq$ negation of one part.

The Kneales identify this discussion as the first appearance in the history of logic of something recognizably like the notion of the **scope of an operator**.

38 Double Negation

The Stoics also recognize a double negative and hold it equivalent to the corresponding affirmative.
In modern notation:

$$\neg\neg P$$

is equivalent to:

$$P.$$

DOUBLE NEGATION

$$\neg\neg P \iff P$$

Stoic terminology includes the idea of a “super-negative.”

This principle later becomes important in the reconstruction of Stoic derivations.

39 Universal Statements as Conditionals

Another striking difference from Aristotle appears in the Stoic treatment of general propositions.
A statement such as:

Man is a rational mortal animal

can be represented approximately as:

If anything is a man, then it is a rational mortal animal.

Thus universality is expressed by conditional structure.

ARISTOTELIAN STYLE

Every man is mortal.

STOIC-STYLE ANALYSIS

If anything is a man, then it is mortal.

universal predication $\rightarrow$ conditional structure

This helps explain why conditionals become so central in Stoic logic.

40 Simple and Complex Axiomata

The Stoics distinguish simple from non-simple or complex axiomata.

Complex axiomata are constructed by means of connectives.

The ancient classification includes several forms.

For our logical purposes the principal structures can be displayed using modern notation.

COMPLEX AXIOMATA

$\neg P$	=	negation
$P \wedge Q$	=	conjunction
$P \vee Q$	=	disjunction
$P \rightarrow Q$	=	conditional
$\neg(P \wedge Q)$	=	“not both”

Modern symbols are ours, not Stoic notation.

The Stoic classification also includes causal and comparative complexes, but these need not occupy our main formal discussion.

41 Conjunction

The Stoics recognized the familiar truth condition for conjunction.

A conjunction is true when all of its components are true and false when at least one component is false.

CONJUNCTION

$$P \wedge Q$$

is TRUE iff:

$$P = T \quad \text{and} \quad Q = T$$

Otherwise:

$$P \wedge Q = F$$

This is a clear example of truth-functional treatment.

42 Stoic Disjunction

The early Stoic disjunction is normally strong or exclusive.

Thus:

$$P \text{ XOR } Q$$

means:

one or the other, but not both.

Later Stoics also recognized a weaker form under which the disjunction is true if at least one disjunct is true.

STOIC “OR”

Early strong form:

$$P \text{ XOR } Q$$

means:

one OR the other, BUT NOT BOTH

Later tradition also recognizes:

$$P \vee Q$$

where both may be true.

This distinction is important because Chrysippus’ fourth and fifth indemonstrables depend upon the strong disjunction.

43 Axioma and the Modern Proposition

It is tempting to identify:

axioma = proposition.

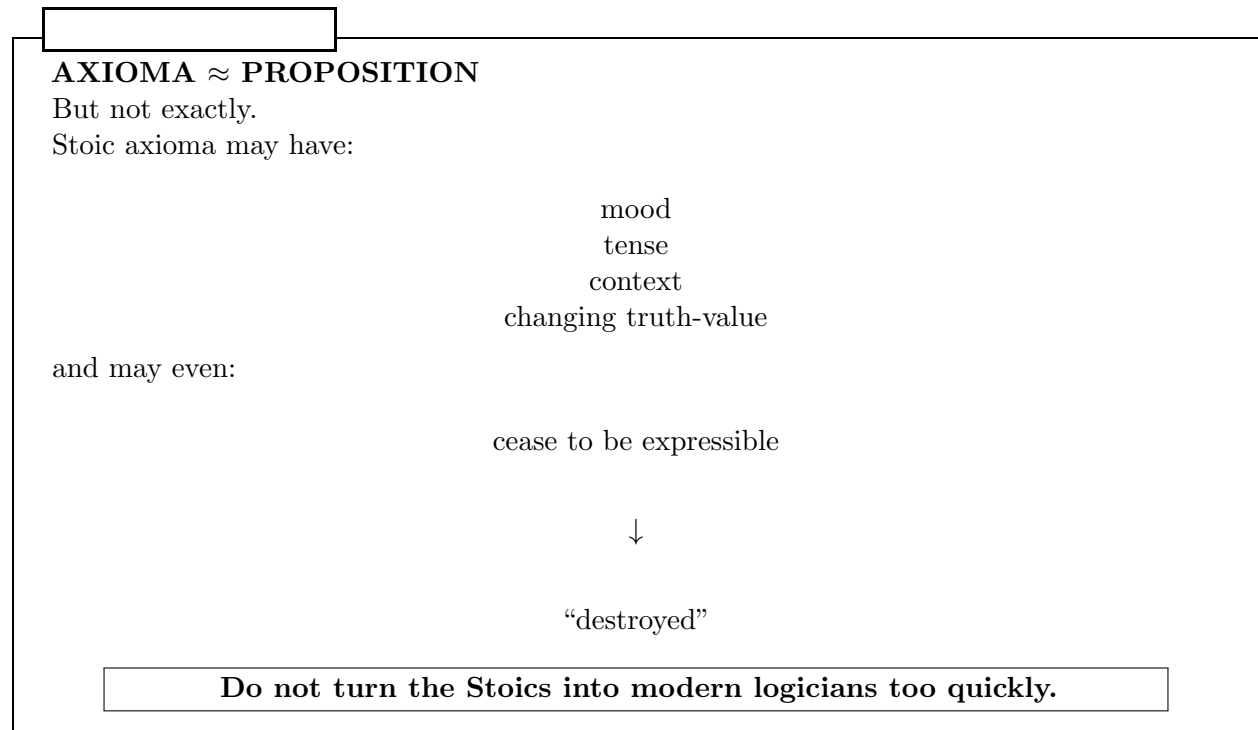
The comparison is useful but not exact.

Stoic axiomata can possess features that modern philosophers often exclude from abstract propositions.

They can involve:

- mood;
- tense;
- context-sensitive reference;
- changing truth-values.

Some Stoic discussions even allow an axiom to be “destroyed” when the mode of reference upon which it depends can no longer operate.



The Stoics have clearly distinguished linguistic expression from what it means, but their lekta preserve more features of utterance-context than many later theories of propositions do.

V. The Stoic System of Inference Schemata

44 The Fundamental Contrast with Aristotle

We have now reached the formal center of Stoic logic.

The key contrast with Aristotle concerns the kinds of variables used.

Aristotle uses letters as places for **terms**.

The Stoics instead use ordinal expressions such as:

the first, the second, the third

as places for **whole propositions**.

This difference is fundamental.

ARISTOTLE

$$\frac{\text{Every } M \text{ is } L \\ \text{Every } S \text{ is } M}{\therefore \text{Every } S \text{ is } L}$$

$M, S, L \rightarrow$ TERM VARIABLES

STOICS

$$\frac{\text{If the FIRST, then the SECOND} \\ \text{The FIRST}}{\therefore \text{The SECOND}}$$

$P, Q \rightarrow$ PROPOSITIONAL VARIABLES

This is the key to understanding why Stoic logic anticipates what later came to be called propositional logic.

45 Inference Schemata

Where Aristotle often formulates a logical principle as a conditional statement about terms, the Stoics regularly present **skeleton arguments**.

Using modern letters merely for convenience:

INFERENCE SCHEMA

$$\frac{P \rightarrow Q \\ P}{Q}$$

$P, Q =$ whole propositions

The form is a pattern of ARGUMENT, not merely a relation among terms.

The Stoics themselves used ordinal expressions rather than our letters P, Q, R .

46 The Five Indemonstrables

Chrysippus recognizes five fundamental inference schemata.

Ancient authors call them the **indemonstrable moods**.

They function as basic inference patterns from which more complicated arguments can be analyzed.

Because the board is small, they should be written in two packets.

46.1 First and Second Indemonstrables

FIVE INDEMONSTRABLES — I

1.

$$\frac{P \rightarrow Q \quad P}{Q}$$

2.

$$\frac{P \rightarrow Q \quad \neg Q}{\neg P}$$

Modern later names:

1 = modus ponens

2 = modus tollens

Copy, then erase.

The Latin names are later.

They are useful labels for us, but they should not be projected back onto Chrysippus.

46.2 Third, Fourth, and Fifth Indemonstrables

The third uses a “not both” proposition.

The fourth and fifth use the Stoics’ strong or exclusive disjunction.

FIVE INDEMONSTRABLES — II

3.

4.

$$\frac{\neg(P \wedge Q) \quad P}{\neg Q}$$

5.

$$\frac{P \text{ XOR } Q \quad P}{\neg Q}$$

$$\frac{P \text{ XOR } Q \quad \neg Q}{P}$$

XOR = exclusive OR

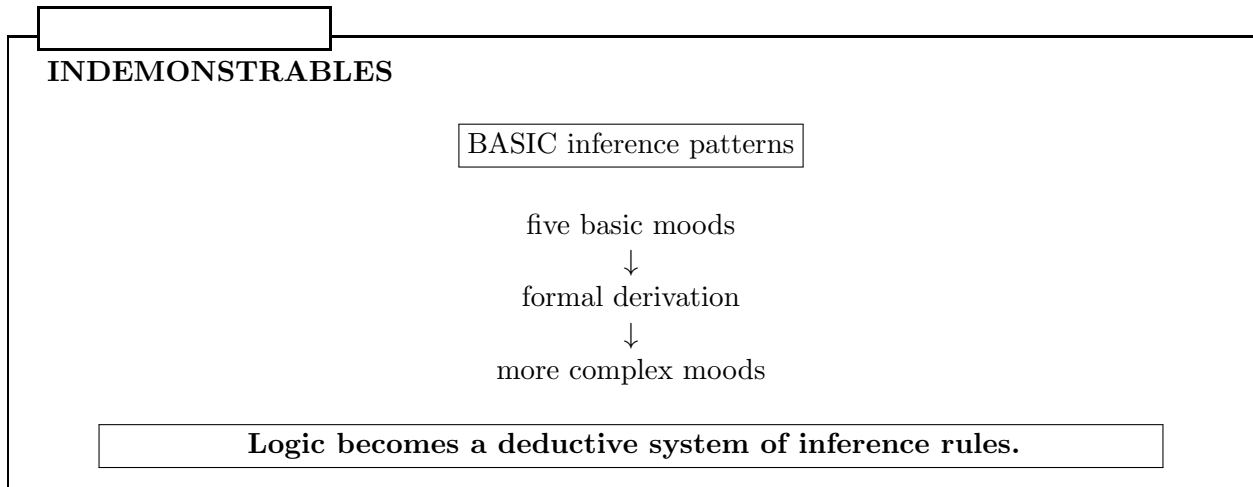
Together these five patterns form the basic machinery of Chrysippean inference.

47 Why They Are Called Indemonstrable

“Indemonstrable” does not mean unintelligible or irrational.

The five forms function as basic inference patterns rather than patterns that must themselves be derived from still more fundamental Stoic moods.

More complicated valid forms can then be analyzed in terms of them.



This is a crucial difference between merely collecting examples of good arguments and constructing a logical system.

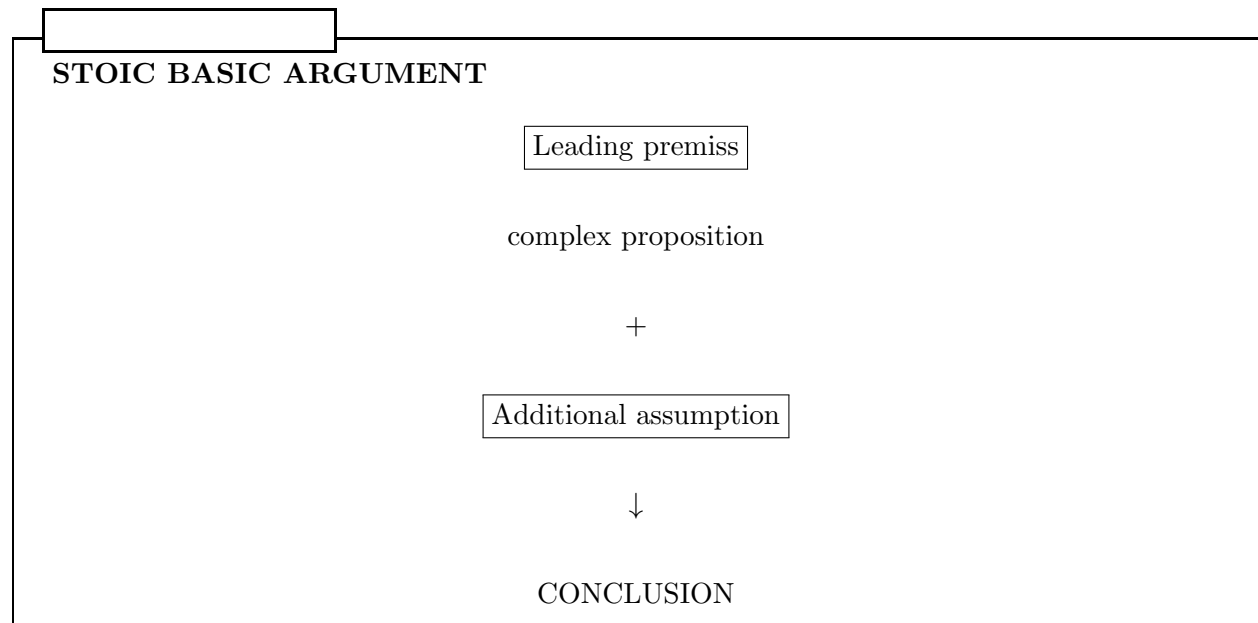
48 Leading Premise and Additional Assumption

For Chrysippus, the simplest argument normally contains two premisses.

One is complex and determines the characteristic structure of the argument.

This is called the **leading premiss**.

The second is the **additional assumption**.



Chrysippus regarded the two-premiss form as basic.

Antipater later recognized one-premiss inference, and this was remembered as an innovation.

	Chrysippus:	simplest inference	= 2 premisses
	Antipater:	one-premiss inference	= allowed

49 Derived Inference

The most important feature of the Stoic system is that complicated arguments can be reduced to sequences of simpler ones.

Consider:

$$(P \wedge Q) \rightarrow R,$$

$$\neg R,$$

$$P.$$

We wish to derive:

$$\neg Q.$$

First use the second indemonstrable on:

$$(P \wedge Q) \rightarrow R$$

and:

$$\neg R.$$

This yields:

$$\neg(P \wedge Q).$$

Then apply the third indemonstrable to:

$$\neg(P \wedge Q)$$

and:

$$P.$$

The result is:

$$\neg Q.$$

Because the board is small, do the derivation in packets.

DERIVED ARGUMENT — SETUP

$$\frac{\begin{array}{c} (P \wedge Q) \rightarrow R \\ \neg R \\ P \end{array}}{\therefore \neg Q}$$

Copy, then erase.

DERIVED ARGUMENT — STEP 1

$$\frac{(P \wedge Q) \rightarrow R \quad \neg R}{\neg(P \wedge Q)}$$

Using Indemonstrable 2.

DERIVED ARGUMENT — STEP 2

$$\frac{\neg(P \wedge Q) \quad P}{\neg Q}$$

Using Indemonstrable 3.

Complex inference = sequence of basic moods

This is one of the clearest surviving examples of the systematic character of Stoic logic.

50 Intermediate Conclusions

The preceding demonstration depends on an important proof principle.

What has been derived from some premisses may itself be used as a premiss in a further inference.

Schematically:

$$P_1, P_2 \vdash Q$$

and:

$$Q, P_3 \vdash R.$$

Then:

$$P_1, P_2, P_3 \vdash R.$$

INTERMEDIATE CONCLUSIONS

$$\frac{\begin{array}{c} P_1, P_2 \vdash Q \\ Q, P_3 \vdash R \end{array}}{P_1, P_2, P_3 \vdash R}$$

An intermediate conclusion may become a new premiss.

This principle is obvious to a modern student familiar with formal proof, but its explicit use is historically significant.

51 The Themata

Ancient reports also tell us that the Stoics used four **themata** in the analysis of complicated arguments.

Unfortunately, only fragments of this doctrine survive.

The first and third are known with reasonable clarity.

The second and fourth are lost and can only be reconstructed conjecturally.

The Kneales therefore treat this part of Stoic logic cautiously.

THEMATA

general rules for ANALYZING complex arguments

Four were recognized.

But:

1st + 3rd survive clearly

2nd + 4th are lost

52 The First Thema

The first thema can be represented approximately as follows.

If two premisses entail a conclusion, then either premiss together with the negation of the conclusion entails the negation of the other premiss.

Schematically:

$$P, Q \vdash R$$

gives:

$$P, \neg R \vdash \neg Q.$$

FIRST THEMA

If:

$$P, Q \vdash R$$

then:

$$P, \neg R \vdash \neg Q$$

and correspondingly:

$$Q, \neg R \vdash \neg P$$

This is closely related to the familiar technique of indirect argument.

53 The Third Thema

The third thema concerns the substitution of a derivation into a larger argument.

Suppose some argument requires Q as a premiss.

If Q has itself been established from P_1 and P_2 , those supporting premisses can replace Q in the larger derivation.

THIRD THEMA

Suppose:

$$P_1, P_2 \vdash Q$$

and:

$$Q, R \vdash S.$$

Then:

$$P_1, P_2, R \vdash S$$

derived premiss $\rightarrow$ replace by its supporting premisses

The principle helps explain how the Stoics could reduce complicated arguments to combinations of their basic forms.

54 Conditionalization

The Kneales' reconstruction of Stoic procedure suggests another important principle.

If:

$$P, Q \vdash R,$$

the corresponding necessary connexion can be represented by the conditional:

$$(P \wedge Q) \rightarrow R.$$

This allows a valid inference relation to be introduced as a conditional premiss inside another argument.

CONDITIONALIZATION	
	$P, Q \vdash R$
becomes:	
	$(P \wedge Q) \rightarrow R$
	inference pattern $\rightarrow$ conditional premiss

This is an important modern reconstruction of how the Stoic system may have operated.

It should not be presented as though this exact symbolic rule survived in Chrysippus' own writings.

55 Reductio Returns

One of the most important Stoic derived patterns brings us back to the very origins of the Megarian tradition.

Suppose:

$$P \rightarrow Q$$

and:

$$P \rightarrow \neg Q.$$

Then:

$$\neg P.$$

STOIC DERIVED REDUCTIO

$$\frac{P \rightarrow Q \quad P \rightarrow \neg Q}{\neg P}$$

Historical line:

Zeno $\rightarrow$ Megarians $\rightarrow$ Chrysippus

Dialectical practice has become formal theory.

This is historically beautiful.

The style of argument that had helped stimulate Megarian logic has now been incorporated into a systematic formal calculus.

56 Consequentia Mirabilis

Another striking form associated with Stoic logical development is:

$$\neg P \rightarrow P.$$

From this, under the relevant principles of the system, one may derive:

$$P.$$

The pattern later came to be known as *consequentia mirabilis*.

CONSEQUENTIA MIRABILIS

$$\frac{\neg P \rightarrow P}{P}$$

Interpretive idea:

If denying P forces P , then P .

The pattern became important in later anti-sceptical and logical arguments.

But we should not assume that every philosophical application of the form establishes as much as its user claims.

57 Rigour and Formalism

Ancient Peripatetics sometimes criticized the Stoics for excessive attention to verbal or formal details.

Chrysippus apparently proved even results that other philosophers were content simply to assume.

To modern students trained in mathematical reasoning, this “excessive” concern for form may look much more like a virtue than a defect.

ANCIENT CRITICISM

Stoics:

too formal

too concerned with exact expression

Modern perspective:

formal precision is a major logical achievement

Unfortunately, the hostile ancient reputation of Stoic formalism may have contributed to the disappearance of Chrysippus’ logical works.

58 Did the Stoics Claim Completeness?

Ancient reports suggest that Chrysippus believed complicated valid arguments could ultimately be analyzed through the five indemonstrables.

This resembles a claim of completeness.

But we must be careful.

The Stoics did not possess our modern technical definition of a complete formal calculus, nor do we possess an ancient proof of completeness in the modern sense.

STOIC SYSTEMATIC CLAIM

valid arguments

↓

analysis



FIVE INDEMONSTRABLES

But:

ancient “complete” $\neq$ modern completeness theorem

Modern reconstructions can prove strong results for reconstructed Stoic systems, but those achievements must not simply be attributed to Chrysippus without evidence.

59 Aristotle and Chrysippus Compared

We can now make the central comparison of the chapter.

Aristotle’s variables range over **general terms**.

Stoic variables range over **propositions**.

Aristotle therefore analyzes logical relations involving the internal structure of propositions.

The Stoics analyze patterns of inference determined by the ways whole propositions are joined by connectives.

ARISTOTLE

$$\begin{array}{l} \text{Every } M \text{ is } L \\ \text{Every } S \text{ is } M \\ \hline \text{Every } S \text{ is } L \end{array}$$

variables $\rightarrow$ TERMS

CHRYSIPPUS

$$\begin{array}{l} P \rightarrow Q \\ P \\ \hline Q \end{array}$$

variables $\rightarrow$ PROPOSITIONS

The distinction can be stated in one line.

	Aristotle → relations WITHIN propositions Stoics → relations BETWEEN propositions

This is why the two systems are complementary rather than genuine competitors.

60 Primary Logic

The Kneales propose the term **primary logic** for the fundamental theory involving such notions as:

$$\neg, \quad \wedge, \quad \vee, \quad \rightarrow.$$

This logic deals with the combinations and inferential relations of propositions without yet introducing the generality expressed by “every” and “some.”

	PRIMARY LOGIC
	$\neg$ $\wedge$ $\vee$ $\rightarrow$
Concern:	
	relations among propositions

Stoic dialectic is the first substantial ancient version of this kind of logic.

61 General Logic

The Kneales reserve the expression **general logic** for the larger theory in which primary logic is combined with notions of generality such as:

every

and:

some.

Thus general logic includes quantificational structure.

	GENERAL LOGIC

PRIMARY LOGIC + EVERY / SOME

That is:

connectives + quantification

Primary logic is more fundamental not because it contains general logic, but because general logic presupposes it.

General logic presupposes primary logic.

But:

primary logic can be developed independently.

62 Where Aristotle and Chrysippus Fit

With this terminology, the Kneales give a striking reinterpretation of ancient logic.

Aristotle’s syllogistic is a **fragment of general logic**.

It deals explicitly with generality and relations among general terms.

But it tacitly relies upon principles of primary logic that Aristotle does not formulate as a separate system.

Chrysippus, by contrast, produces the first major ancient **primary logic**.

TWO ACHIEVEMENTS

Chrysippus → first primary logic

Aristotle → fragment of general logic

Aristotle:

quantification without a separate

formal calculus of propositional connexion

Chrysippus:

propositional connexion without

Aristotle’s quantificational term theory

The two systems solve different parts of the larger problem.

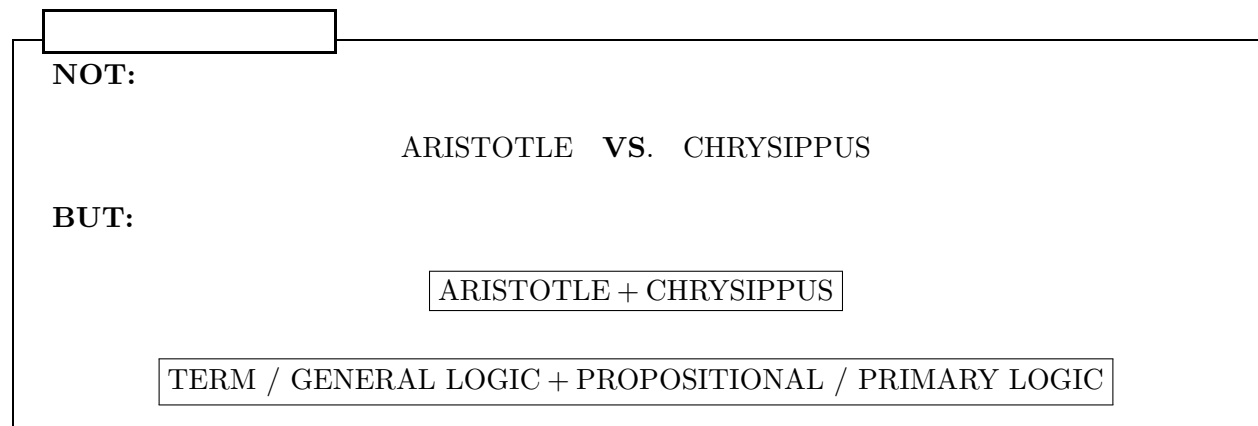
63 The Correct Historical Picture

Ancient partisans tended to ask:

Aristotle or Chrysippus?

The more fruitful question is:

How do the two systems fit together?



The intellectual loss of antiquity was not that one school chose the wrong logic.

It was that the complementary character of the two achievements was obscured by rivalry.

64 Chapter Summary

The Megarian–Stoic tradition begins from a different source than Aristotle’s logic.

Instead of taking geometrical demonstration as its principal model, it grows from dialectic, reductio, paradox, and argument with complex propositions.

Eubulides’ paradoxes expose problems of truth, identity, vagueness, and presupposition.

Diodorus and Philo develop rival accounts of possibility.

Diodorus’ Master Argument attempts to tie possibility to what is or will be true, but the Kneales argue that the reasoning depends upon confusion between sentences, tense, and the contents those sentences express.

The Megarian debate over conditionals then asks what makes an “if ... then ...” proposition sound.

Philo develops a truth-functional account.

Diodorus proposes a stronger temporally constrained account.

The theory of connexion requires incompatibility between the antecedent and the negation of the consequent.

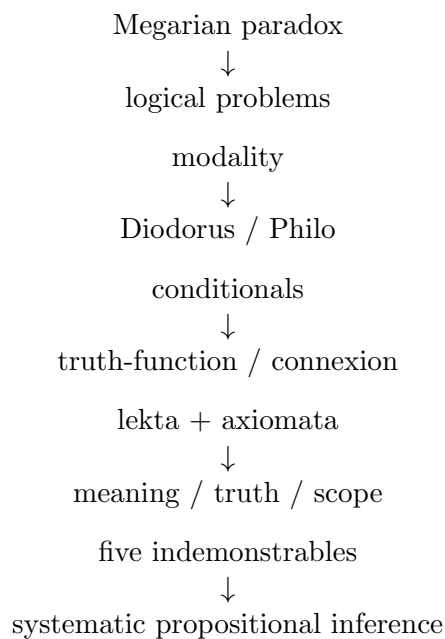
The Stoics then develop a sophisticated semantic theory distinguishing expressions, objects, and incorporeal lekta.

Among complete lekta, axiomata are the principal bearers of truth and falsity.

Stoic discussion of negation introduces an explicit concern with the scope of an operator.

Finally, Chrysippus systematizes inference among whole propositions through five indemonstrable moods and procedures for deriving more complex arguments.

CHAPTER III SUMMARY



65 Final Historical Takeaway

The great achievement of Aristotle was the systematic study of inferences involving general terms.

The great achievement of the Stoics was the systematic study of inferences whose validity depends upon the logical relations among whole propositions.

Neither achievement eliminates the other.

A mature logic requires both.

FINAL TAKEAWAY

ARISTOTLE

terms
syllogistic
general logic

CHRYSIPPUS

propositions
connectives
primary logic

Ancient logic discovered two major pieces
of one larger logical system.

Next: Roman and Medieval Logic